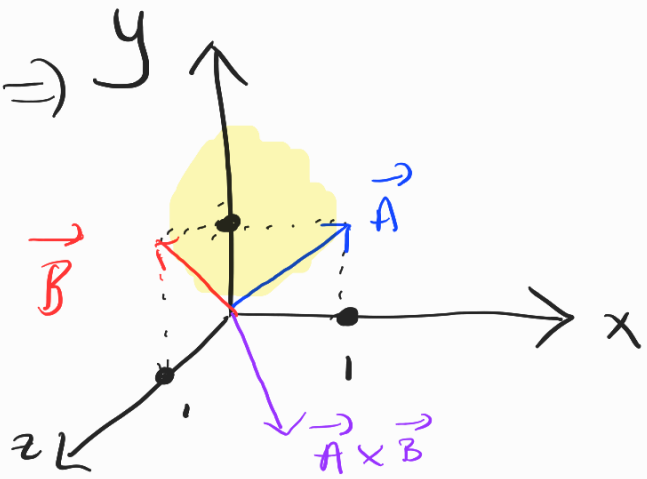
(1) $\vec{A} = \hat{i} + \hat{j} = (1,1,0)$

$\vec{B} = \hat{j} + \hat{k} = (0,1,1)$

$\Rightarrow$

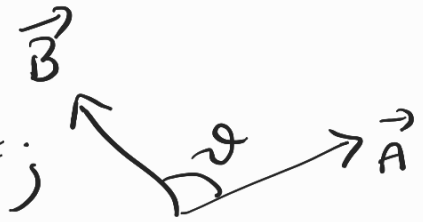


(i) $\vec{A}\cdot\vec{B} = (1\cdot 0) + (1\cdot 1) + (0\cdot 1) = 1$ //

(ii) $\vec{A}\times\vec{B} = \begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \hat{i}(1-0) - \hat{j}(1-0) + \hat{k}(1-0)$

$= \hat{i} - \hat{j} + \hat{k}$ //

(iii)



$|A| = \sqrt{1^2+1^2} = \sqrt{2}$

$|B| = \sqrt{2}$

$\vec{A}\cdot\vec{B} = |A||B|\cos\vartheta = 1$

$\Rightarrow \sqrt{2}\cdot\sqrt{2}\cos\vartheta = 1$

$\Rightarrow \cos\vartheta = 1/2 \Rightarrow \vartheta = \arccos(1/2)$

$\Rightarrow \vartheta = 60^\circ$

(iv) κάθετο $\Rightarrow \vec{A}\times\vec{B} = \vec{C}$

$\Rightarrow \hat{C} = \dfrac{\vec{C}}{|C|} = \dfrac{(1,-1,1)}{\sqrt{1+1+1}} = \dfrac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$ //

② $\vec{r} = 30t\,\hat{i} + (40t - 5t^2)\hat{j}$

$\vec{v}(t) = ?$, $\vec{a}(t) = ?$

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = 30\hat{i} + (40 - 10t)\hat{j} \;//$$

$$\vec{a} = \frac{d}{dt}\vec{v}(t) = 0\hat{i} - 10\hat{j} = -10\hat{j} \;//$$

(3)

$$x = v_0 \cos\vartheta\, t \quad (1)$$

$$y = v_0 \sin\vartheta\, t - \tfrac{1}{2} g t^2 \quad (2)$$

$$(1): t = \frac{x}{v_0 \cos\vartheta} \quad (3)$$

$$(3) \rightarrow (2):$$

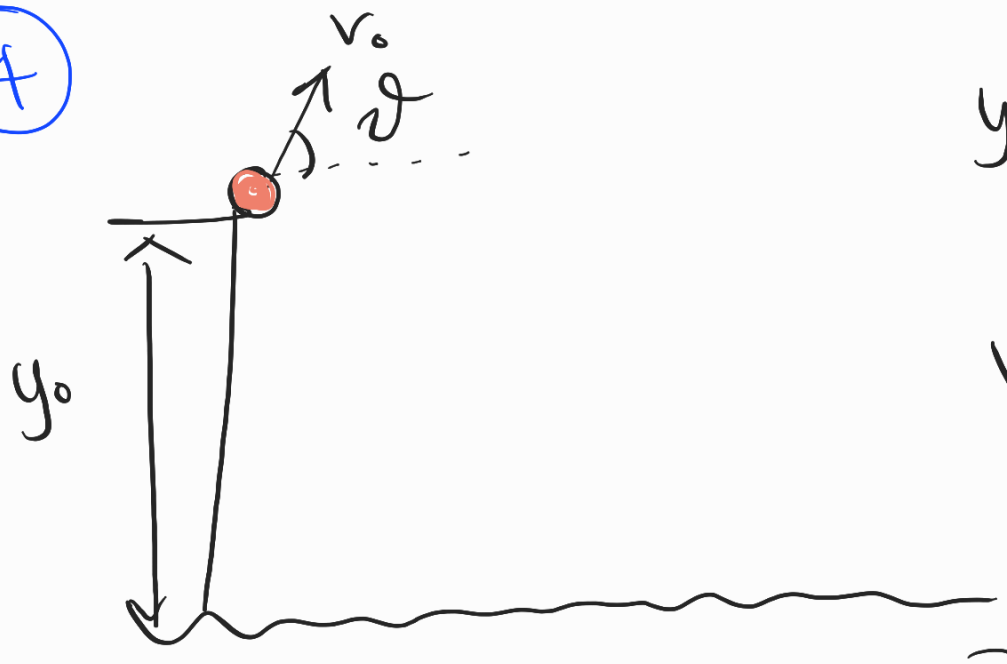
$$y = \frac{v_0 \sin\vartheta \cdot x}{v_0 \cos\vartheta} - \frac{1}{2} g \left(\frac{x}{v_0 \cos\vartheta}\right)^2$$

$$= x \tan\vartheta - \frac{g x^2}{2 v_0^2} \cdot \frac{1}{\cos^2\vartheta}$$

$$\rightarrow \sec^2\vartheta$$

$$\Rightarrow y = x \tan\vartheta - \frac{1}{2} \frac{g x^2}{v_0^2} \sec^2\vartheta$$

4



$y_0 = 150m$

$\vartheta = 60°$

$v_0 = 100 m/s$

(i) $\vec{r}(t) = ?$, $\vec{v}(t) = ?$

$$\vec{r} = r_x \hat{i} + r_y \hat{j}$$

$$r_x : x = v_0 \cos\vartheta t$$

$$r_y : y = y_0 + v_0 \sin\vartheta t - \frac{1}{2} g t^2$$

$$\Rightarrow \vec{r}(t) = (v_0 \cos\vartheta t)\hat{i} + (y_0 + v_0 \sin\vartheta t - \tfrac{1}{2} g t^2)\hat{j}$$

$$\Rightarrow \vec{r}(t) = (100 \cdot \cos(60) t)\hat{i} + (150 + 100 \sin(60) t - 5t^2)\hat{j}$$

$$\Rightarrow \vec{r}(t) = 50t\hat{i} + (150 + 87t - 5t^2)\hat{j} \ m$$

$$\vec{v}(t) = \frac{d}{dt}(\vec{r}(t))$$

$$= 50\hat{i} + (87 - 10t)\hat{j} \text{ m/s}$$

(ii) Διαφορά $\Rightarrow y = 0$

$$y = r_y(t) \Rightarrow y = 150 + 87t - 5t^2$$

$$\Rightarrow 5t^2 - 87t - 150 = 0$$

$$\Rightarrow t = -1.6\,s$$

~~$t = -1.6\,s$~~ απ

$$\boxed{t = 19\,s}$$

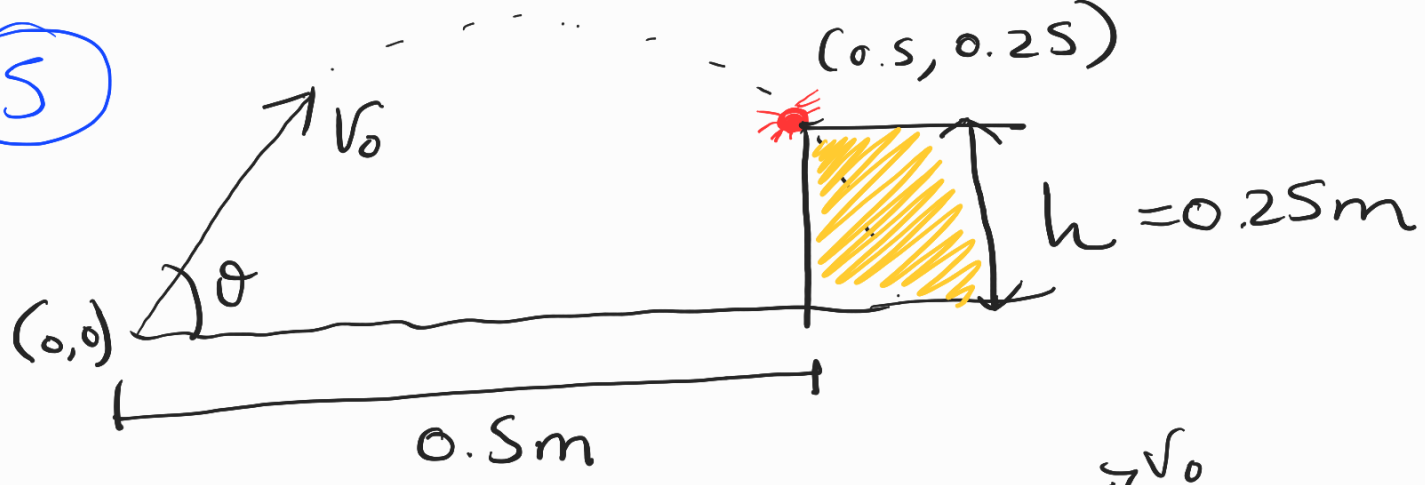
για $t = 19\,s$: $\vec{r} = 50(19)\hat{i} + (150 + 87 \cdot 19 - 5(19)^2)\hat{j}$

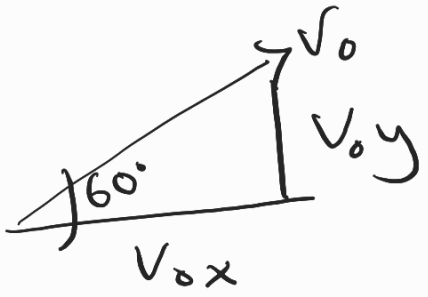
$$\Rightarrow \boxed{\vec{r} = 950\hat{i} + 2\hat{j}\ m}$$

$$\vec{v} = 50\hat{i} + (87 - 190)\hat{j}$$

$$\boxed{\vec{v} = 50\hat{i} - 103\hat{j}\ m/s}$$

(5)





(i) $v_0 = ?$

$$x = x_0 + v_{0x} t$$

$$y = y_0 + v_{0y} t - \frac{1}{2} g t^2$$

$$\Rightarrow x = x_0 + v_0 \cos\theta \cdot t \quad ①$$

$$y = y_0 + v_0 \sin\theta\, t - \tfrac{1}{2} g t^2 \quad ②$$

$$① : \frac{x - x_0}{v_0 \cos\theta} = t \quad ③$$

$$③ \to ② : y = y_0 + \frac{v_0 \sin\theta\,(x - x_0)}{v_0 \cos\theta} - \frac{1}{2} g \left(\frac{x - x_0}{v_0 \cos\theta}\right)^2$$

$\sin\theta / \cos\theta \to \tan\theta$

$y_0 = 0, x_0 = 0$

$y = 0.25, x = 0.5$

$\theta = 60°$

$$\Rightarrow 0.25 = 0 + \tan(60^\circ)(0.5) - 5 \cdot \frac{0.5^2}{V_0^2 \cos^2(60)}$$

$$\Rightarrow 0.25 = (1.73)(0.5) - \frac{5 \cdot (0.5)^2}{V_0^2 (0.5)^2}$$

$$\Rightarrow 0.25 = 0.87 - 5/V_0^2$$

$$\Rightarrow 5/V_0^2 = 0.62 \Rightarrow V_0^2 = 5/0.62$$

$$\Rightarrow V_0 = 2.84 \text{ m/s} \;//$$

(ii) $V_y = \pm ?$

$\Rightarrow$ find V_y at spider

$$\Rightarrow V_y = V_{oy} - gt = V_0 \sin\theta - gt \quad ④$$

$$③: t = \frac{x}{V_0 \cos\theta}$$

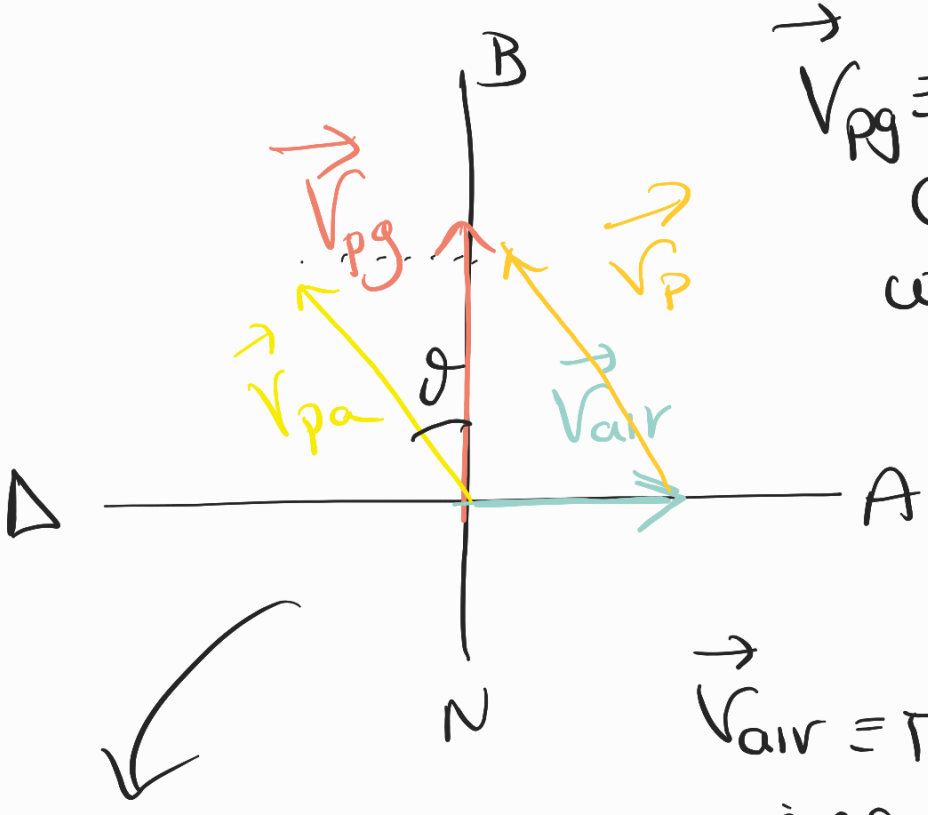
$$③ \rightarrow ④: V_y = (2.84)\sin(60^\circ) - \frac{10 \cdot (0.5)}{2.84 \cos(60)}$$

$$\Rightarrow V_y = (2.84)(0.87) - \frac{10(0.5)}{(2.84)(0.5)}$$

$$= 2.47 - 3.52$$

$$= -1.05 \text{ m/s} \Rightarrow \text{waterbairet} //$$

6



$\vec{V}_{pg}$ ≡ ταχύτητα αεροπλάνου ως προς το έδαφος

$\vec{V}_{air}$ ≡ ταχύτητα αέρα ως προς έδαφος

$\vec{V}_{pa}$ ≡ ταχύτητα αεροπλάνου ως προς τον αέρα

$$\vec{V}_{pa} + \vec{V}_{air} = \vec{V}_{pg}$$

$$V_{pa} = 200\ km/h$$

$$V_{air} = 90\ km/h$$

$$\Rightarrow \frac{V_{air}}{V_a} = \sin\vartheta \Rightarrow \frac{90}{200} = \sin\vartheta$$

$$\Rightarrow \vartheta = 26.7^\circ$$

(ii) $\frac{V_{pg}}{V_{pa}} = \cos\vartheta$

$\Rightarrow V_{pg} = 200 \cdot \cos(26.7)$

$= 178.6 \text{ km/h}$ //